

$$MA1 = 50$$

$$1. \quad |Z_n| = 12$$

$$|U(12)| = 4$$

$$|U(12)| = 4$$

$$|U(20)| = 8$$

$$|D_4| = 8$$

The orders of elements in groups are a factor of the order of the group.

$$2 \quad \langle 0.5 \rangle = \{0.5n \mid n \in \mathbb{Z}\}$$

$$\langle 0.5 \rangle = \{\dots, -2.0, -1.5, -1.0, -0.5, 0, 0.5, 1.0, \dots\} \text{ under } \mathbb{Q}^+$$

$$\langle 0.5 \rangle = \{0.5^n \mid n \in \mathbb{Z}\}$$

$$\langle 0.5 \rangle = \{\dots, 8, 4, 2, 1, 0.5, 0.25, 0.125, \dots\} \text{ under } \mathbb{Q}^*$$

3 Since the subgroup possesses the same operation as $\mathbb{Q}^+$ no non-zero element in the subgroup has finite order because rational $\neq$ under addition are infinite and there's no $n \neq 0$ that will give 0 when multiplied by a non-zero rational $\neq$.
Order of every non-zero element in $\langle 0.5 \rangle$ is infinite.
 $|0| = 1$ since $1 \cdot 0 = 0$

likewise in the subgroup under $\mathbb{Q}^*$ generated by $\frac{1}{2}$, all powers $(\frac{1}{2})^n$ for $n \in \mathbb{Z}$ are distinct non-zero numbers, so all elements except identity 1, have infinite orders.
 $|1| = 1$ since $1 \cdot 1 = 1$

4 Let G be a group and $g \in G$ and $|g| = n$

This means $g^n = e$ (where $|g^n| = n : (g^n)^n = e$)

$$g^n = e$$

$$(g^n)^n = e^n = e$$

$$(g^n)^n = e$$

$$\therefore |g^n| = n$$

7. $a^6 = e$

$b^7 = e$

Using property of inverses, we know $(a^4 c^2 b^4)^{-1} = (b^4)^{-1} (c^2)^{-1} (a^4)^{-1}$
Simplifying each part we get

$(b^4)^{-1} = b^{-4}$

Since $|b| = 7$; $b^{-4} = e b^{-4} = b^7 b^{-4} = b^{7-4} = b^3$

$(c^2)^{-1} = c^{-2}$

$(a^4)^{-1} = a^{-4}$

since $|a| = 6$, $a^{-4} = e a^{-4} = a^6 a^{-4} = a^{6-4} = a^2$

$\therefore (a^4 c^2 b^4)^{-1} = b^3 c^2 a^2$

6. In D_8 , a combination of a flip and rotation yields another flip. Moreover, flip with flip will yield rotation, while rotation combined with rotation is another rotation.

Therefore, the subgroup of D_8 containing R_{240} and reflection F , must be the entire group D_8 .

i. Let F_i and F_j be 2 elements of a subgroup of D_8 .

The square of any of the reflection is the identity R_0 .

Moreover, $F_i \cdot F_j = R_k$ for some rotation.

Thus, when a subgroup contains 2 different reflections. Composition of those reflections are rotations, and those rotations are equally accompanied with their inverses.

Hence all rotations must be in the subgroup; however, the 3rd reflection, F_k , is independent of the two reflections already given in the prompt, and so doesn't necessarily show up. Unlike in (i), a subgroup containing 2 reflections is not necessarily the entire group D_8 because the third reflection is omitted in this case.

Subgroup containing reflection F_i & $F_j = \{R_0, R_{120}, R_{240}, F_i, F_j\}$

9. Let R_{z70} and V be in a subgroup of D_4 .

Since $R_{z70} \in$ subgroup of D_4 , we know $(R_{z70})^{-1}$ must also exist

From Cayley table $(R_{z70})^{-1} = R_{z270}$

Additionally, we know a reflection is its own inverse, so

$$V \cdot V = e = R_0$$

So the identity element is in the subgroup

Moreover, the composition of a rotation & reflection in D_4 yields another reflection.

So $V \cdot R_0$; $V \cdot R_{90}$; $V \cdot R_{180}$ are all unique reflections in D_4

however $V \cdot R_0 = R_0 \cdot V = V$

$\therefore$ A subgroup containing R_{z70} & reflection V for instance, looks like $\{R_0, R_{90}, R_{z70}, V, D, D'\}$

ii. Let H and D be in a subgroup of D_4 .

We know composition of 2 reflections yields a rotation.

Since every reflective symmetry in D_4 is not commutative

$$H \cdot D \neq D \cdot H$$

Also the reflections are their own inverses so $H \cdot H = D \cdot D = e = R_0$
 R_0 must be in the subgroup.

Now, we know that the composition of 2 unique reflections in D_4 cannot map to R_0 but to only another ordered set of rotations we can generalize as $\{R_0, R_{180}\}$

In this case, $R_{180} = H \cdot D = R_{180}$

$$R_{180} = D \cdot H = R_{180}$$

$\therefore$ If a subgroup of D_4 contains H and D , it must contain at least 5 elements including identity: $\{R_0, R_{180}, R_{z70}, H, D\}$

iii. Let H and V be in a subgroup of D_4 .

Since H and V commute we know $H \cdot V = V \cdot H = R_{z90}$ must be also in the subgroup above.

We know in a dihedral group $n=4$ $R_{z90} \in Z(D_4)$

This means $H \cdot R_{z90} = R_{z90} \cdot H = V$; $V \cdot R_{z90} = R_{z90} \cdot V = H$

Which turn out to be "multiplicative inverses of each other"

$\therefore$ Subgroup of D_4 containing H and $V = \{R_0, R_{180}, H, V\}$
where $R_0 = V' = H''$

12. $|x| = 2$
 $x^2 = x \cdot x = e$

13. Let G be a group and $a, x \in G$ and let $|a| = n$
[show $|xax^{-1}| = n$]

This means $a^n = e$ [show $(xax^{-1})^n = e$]

Expanding, we have $(xax^{-1})^n = (xax^{-1})(xax^{-1})(xax^{-1}) \dots (n \text{ times})$

By associativity $(xax^{-1})^n = xax^{-1}xax^{-1}xax^{-1} \dots (n \text{ times})$
 $= xa^n x^{-1}$

Replacing $a^n = e$ above we get

$(xax^{-1})^n = xex^{-1} = e$ because $e \in Z(G)$

14. Let G be a group and $a \in G$

From the problem, we have $|a| = 2$ is the only element of order 2 i.e. $a^2 = e$ [show $a \in Z(G)$]

Let $g \in G$. Consider gaq^{-1} [show $gaq^{-1} = a$]

which would inherently imply $aq = qa$ for $\forall g \in G$

Since a has order of 2, we will do algebra on gaq^{-1}

$(gaq^{-1})^2 = (gaq^{-1})(gaq^{-1}) = ga(qq)aq^{-1} = gaqaq^{-1} \quad \text{--- (i)}$

Replace $a^2 = e$ in (i)

$(gaq^{-1})^2 = gaaq^{-1} = e = a^2$

This means order of gaq^{-1} is 2. Although, we are given a is the only element of order 2, this means gaq^{-1} must equal a

We get $gaq^{-1} = a$

$gaq^{-1} \cdot q = aq$

$ga = aq$

$\therefore$ a commutes with every element $g \in G$, meaning a must be in $Z(G)$

14. Let G be a group and $a \in G$

Let $|a| = 6$. This means $a^6 = e$

$$|a| = \{n : 6|n, \forall n \in \mathbb{Z}^+\}$$

This is the congruence class of $n \equiv 0 \pmod{6}$

$\therefore$ The possible orders of a must be a multiple of 6, $n = 6K$ for some positive integer K because by definition $|a| \geq 1$

22. Show $U(14)$ can be generated by both 3 and 5 in modulo 14

$$U(14) = \{1, 3, 5, 9, 11, 13\}$$

$$\text{Show } \langle 3 \rangle = U(14), \quad \langle 5 \rangle = U(14)$$

Calculating powers of 3:

$$3^1 \equiv 3 \pmod{14}$$

$$3^2 \equiv 9 \pmod{14}$$

$$3^3 \equiv 27 \equiv 13 \pmod{14}$$

$$3^4 \equiv 3 \cdot 13 = 39 \equiv 11 \pmod{14}$$

$$3^5 \equiv 3 \cdot 11 = 33 \equiv 5 \pmod{14}$$

$$3^6 \equiv 3 \cdot 5 = 15 \equiv 1 \pmod{14}$$

So powers of 3 generate $\{1, 3, 5, 9, 11, 13\} = U(14)$

Again calculating powers of 5:

$$5^1 \equiv 5 \pmod{14}$$

$$5^2 \equiv 25 \equiv 11 \pmod{14}$$

$$5^3 \equiv 5 \cdot 11 = 55 \equiv 13 \pmod{14}$$

$$5^4 \equiv 5 \cdot 13 = 65 \equiv 9 \pmod{14}$$

$$5^5 \equiv 5 \cdot 9 = 45 \equiv 3 \pmod{14}$$

$$5^6 \equiv 5 \cdot 3 = 15 \equiv 1 \pmod{14}$$

So powers of 5 generate the elements $\{1, 3, 5, 9, 11, 13\} = U(14)$

$\therefore$ Hence $U(14)$ is cyclic

b. Find powers of 11 modulo 14

$$11^1 \equiv 11 \pmod{14}$$

$$11^2 \equiv 11 \cdot 11 = 121 \equiv 9 \pmod{14}$$

$$11^3 \equiv 11 \cdot 9 = 99 \equiv 1 \pmod{14}$$

$$\langle 11 \rangle = \{1, 9, 11\} \subset U(14)$$

$$\dots \neq U(14)$$

34 Let G be a group and H, K be subgroups of G [show $H \cap K$ is a subgroup]

(1) non empty: since H and K are subgroups of G , they both contain the identity element of group, G .

That is $e \in H$ and $e \in K$. Hence, $e \in H \cap K$

Thus $H \cap K \neq \emptyset$

(2) closure: Let $x, y \in H \cap K$. This means $x \in H$ and $x \in K$, and $y \in H$ and $y \in K$

Since H and K are subgroups of G , they are both closed under the operation of the group. Thus $xy \in H$ and $xy \in K$

Thus xy must be in $H \cap K$

(3) inverses: Let $x \in H \cap K$. This means $x \in H$ and $x \in K$.

Since we know H & K are subgroups of G , they both contain inverses of every element in their order. Thus $x^{-1} \in H$ and $x^{-1} \in K$

$\therefore x^{-1}$ must be in $H \cap K$ and $H \cap K$ is a subgroup of G

35 Let G be an abelian group and $H = \{x \in G \mid |x| \text{ is odd}\}$
[show H is a subgroup of G]

(1) non empty: In any group G , the identity element has order of 1 and 1 is odd. Thus, $e \in H$

(2) closure: Let $x, y \in H$. This means $|x| = m$, $|y| = n$ where m & n are odd. [show $xy \in H$]

$$m = 2a + 1 \quad \forall a \in \mathbb{N}$$

$$n = 2b + 1 \quad \forall b \in \mathbb{N}$$

$$x \cdot y = (2a+1)(2b+1) = 4ab + 2a + 2b + 1$$

$$= 2(2ab + a + b) + 1 \quad \text{is odd where}$$

$$2ab + a + b \in \mathbb{Z}$$

$$\therefore xy \in H$$

(3) inverses: Let $x \in H$ and $|x| = p$, where p is odd. [show $|x^{-1}|$ is odd]
This means $x^p = e$ [show $|x^{-1}| = p$]

$$(x^p)^{-1} = e^{-1} = e$$

$$(x^p)^{-1} = (x^{-1})^p = e$$

$\therefore |x^{-1}| = |x| = p$ and $x^{-1} \in H$ and H is a subgroup of G

Theorem:

1. In the space below, rewrite the proof as it appears in the text. Then highlight the text using yellow if the statement follows by definition, blue if it is an assumption, green if the statement follows from another theorem, and pink if it is a conclusion.

Clearly, $e \in Z(G)$, so $Z(G)$ is non-empty. Now, suppose $a, b \in Z(G)$. Then $(ab)x = a(bx) = a(xb) = axb = xab = x(ab)$ for all x in G ; and therefore, $ab \in Z(G)$.

Next, assume that $a \in Z(G)$. Then we have $ax = xa$ for all x in G . What we want is $a'x = xa'$ for all x in G . Informally, all we need do to obtain the second equation from the first one is simultaneously to bring the a 's across the equals sign:

$$\begin{array}{c} \downarrow \\ ax = xa \\ \downarrow \end{array}$$

becomes $xa' = a'x$. (Be careful here; groups need not to be commutative. The a on the left comes across as a' on the left, and the a on the right comes across as a' on the right.) Formally, the desired equation can be obtained from the original one by multiplying it on the left and right by a' , like so:

$$a'(ax)a' = a'(xa)a'$$

$$(a'a)xa' = a'x(aa')$$

$$exa' = a'xe$$

$$xa' = a'x$$

This shows $a' \in Z(G)$ whenever a is

2. In the space below, write a summary of the proof using no more than 50 words. The best responses are those that succinctly communicate the most important aspects/ideas of the proofs without reproducing portions of the proof.

The proof utilizes very vividly the two step subgroup test by providing the validity of the theorem's 3 axioms.

3. Where is the first place you struggled to understand the proof?

When the english mixed with math became too condensed at the start of the proof.

4. What, if anything, in this proof do you still not understand?

Why the author drops the warning message of commutation, when clearly the problem is dealing with $Z(G)$ which possesses abelian properties?

5. What would strengthen this proof?

Less beating around the bush, and simple & direct algebraic relationships after introducing an axiom needed to be validated

6. What does the author do well in the construction of this proof?

Swiftly, proves to the audience that it is non-empty and closed using basic set theory notations and some adoption of related theorems like inverse and identity properties in here.

- 20 Let G be any group and $a, b \in G$
 Let $|ab| = n$ [show $|ba| = n$]

This means $(ab)^n = e$

Thus $a^{-1}(ab)^n = a^{-1}e$

$$a^{-1}(ab)^n a = e$$

$$a^{-1}(ab)(ab)(ab) \cdots a = e$$

$$a^{-1}a(ba)(ba)(ba) \cdots = e$$

$$e(ba)(ba)(ba) \cdots = e$$

$$(ba)^n = e$$

$$\therefore |ab| = |ba|$$

- 32 Let H be a subgroup

Since H contains 2^{50} and 3^{50} , it must contain any linear combination of these two elements.

However, given 2 & 3 are coprime, $\gcd(2^{50}, 3^{50}) = 1$

Since $1 \in H$, all integer multiples of 1 is also in H

$$\therefore H = \mathbb{Z}$$

42. $\gcd(8, 14) = 2$

$\langle 8, 14 \rangle = 2\mathbb{Z}$, set of all even integers

b. $\gcd(8, 13) = 1$

$\langle 8, 13 \rangle = \mathbb{Z}$, set of all integers

c. $\gcd(6, 15) = 3$

$\langle 6, 15 \rangle = 3\mathbb{Z}$, set of all integers divisible by 3

d. $\langle m, n \rangle = \{a m + b n \mid a, b \in \mathbb{Z}\}$

e. $\gcd(12, 18, 45) = 3$

$\langle 12, 18, 45 \rangle = 3\mathbb{Z}$, set of all multiples of 3

- 47 Let G be an abelian group with identity e

$$H = \{x \in G \mid x^n = e\}$$

non empty: Since $e \in G$, we know $e^n = e$ for any n

Thus $e \in H$

closure: let $x, y \in H$. This means $x^{-1} = e$ and $y^{-1} = e$

[show $xy \in H$] i.e. $(xy)^{-1} = e$

$$\begin{aligned}\text{Since } G \text{ is Abelian } (xy)^{-1} &= x^{-1}y^{-1} \\ &= e \cdot e = e\end{aligned}$$

$\therefore xy \in H$ so H is closed under group operation

inverses: let $x \in H$. This means $x^{-1} = e$ [show $x^{-1} \in H$]

$$\text{i.e. } (x^{-1})^{-1} = e$$

$$\begin{aligned}\text{Since } G \text{ is abelian } (x^{-1})^{-1} &= (x^{-1})^{-1} \\ &= (e)^{-1} = e\end{aligned}$$

$\therefore x^{-1} \in H$, so H is closed under inverses

The set $H = \{x \in G \mid x^{-1} = e\}$ is a subgroup of G by 2-step subgroup test

48. In $U(8)$

a. Choose $a = 5$; $b = 7$

$$|ab| = |57| = |35| = 3$$

b.

49. Let G be a group and $|G| = n$ where n is even

This means $n = 2k$ for some $k \in \mathbb{Z}$

An element $g \in G$ has order 2 iff $g^2 = e$ which implies g is its own inverse.

Here's the segregation:

Has order 1 (only e , the identity)

Has order 2 (self-inverse)

Has order greater than 2 (this case, they pair with respect to inverse)

For elements with order > 2 , every element g has a distinct inverse $g^{-1} \neq g$

Let k be number of elements of order 2

Non-identity elements can be split into those of order 2 AND those of order greater than 2, which pair up with their respective inverses

Thus, total number of elements in G is

$$n = 1 + k + \text{some even number}$$

Here 1 accounts for identity, k is the count of elements of order 2 and remaining is an even number because these distinct higher order elements come in pairs

Since n is even and 1 is odd, the sum ($k + \text{some even \#}$) must also be odd

$\therefore k$ must be odd

In a group of even order, the # of elements of order 2 must be odd

56 choose $a = 2^n$ ($n \rightarrow \infty$)

$$b = -\frac{1}{2^n} \quad (n \rightarrow \infty)$$

$$|a| = |b| = \infty$$

$$|ab| = \left| 2^n - \frac{1}{2^n} \right| = |1| = 2$$

60a $|U(3)| = 2$, $|U(4)| = 2$, $|U(12)| = 4$

b $|U(5)| = 4$, $|U(7)| = 6$, $|U(35)| = 24$

c $|U(4)| = 2$, $|U(5)| = 4$, $|U(20)| = 8$

d $|U(3)| = 2$, $|U(5)| = 4$, $|U(15)| = 8$

For 2 integers coprime $|U(rs)| = |U(r)| \cdot |U(s)|$

62 $|U(4)| = 2$, $|U(10)| = 4$, $|U(40)| = 16$

No, not really. $\gcd(4, 10) \neq 1$, so the relation will not hold

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non empty: The identity matrix $I \in GL(2, \mathbb{R})$ which possesses determinant of $\det(I) = 1 = 2^0$ (an integer power of 2)

Thus $I \in H$

closure: Let $A, B \in H$. This means $\det(A) = 2^x$ and $\det(B) = 2^y$ for some integers x, y

$$\det(AB) = \det(A) \cdot \det(B) = (2^x)(2^y) = 2^{x+y}$$

Since $z = x + y$ is some integer in $\mathbb{Z}$, $\det(AB)$ is an integer power of 2

$\therefore AB \in H$ and H is closed under the operation

inverses: Let $A \in H$. This means $\det(A) = 2^m$

$$\begin{aligned} \text{From linear algebra, we know } \det(A^{-1}) &= \frac{1}{\det(A)} = \frac{1}{2^m} \\ &= 2^{-m} \end{aligned}$$

Since $k = -m$ is some integer in $\mathbb{Z}$, $\det(A^{-1})$ is an integer power of 2

$\therefore A^{-1} \in H$ and H is closed under inverses

Therefore $H = \{A \in GL(2, \mathbb{R}) \mid \det(A) \text{ is integer power of } 2\}$ is a subgroup of $GL(2, \mathbb{R})$ by the two-step subgroup test